

Replication in field ecology: Identifying challenges and proposing solutions

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Abstract

1. Field ecology has been included in a 'replication crisis' that extends across many scientific disciplines. However, the underlying concepts of replication, reproducibility and replicability are not always clearly distinguished, and complicate the identification of best practices. Furthermore, conducting experiments under the high variability of natural field conditions reduces the capacity for replication relative to other biological disciplines working under controlled conditions. Field ecologists are therefore facing a significant challenge in assessing the replicability of their research with implications for overall confidence in study outcomes.
2. Through a review of the literature, we discuss several related aspects of experimental design that can enhance confidence in scientific outcomes. Specifically, we describe *sample replication* (repeat sample), *within-study replication* (repeat experiment) and *between-study replication* (repeat study) and how each can be used within field ecology. Since perfect between-study replication (i.e. direct replication) is generally not possible in field ecology, we suggest more explicit use of conceptual replication would enhance confidence in scientific outcomes. However, such changes require cultural shifts in practice among all participants in the scientific enterprise.
3. We suggest several tangible steps could be taken to improve confidence in ecological research: (a) increase the use of *within-study replication* before publication, (b) increase replicability for aspects that we can control (e.g. pre-register experiments, open data, publish code), (c) divest from novelty as the primary criterion for publication in leading ecological journals and invest in experimental design, (d) be sceptical of contradictory findings from studies testing similar research questions and (e) create a publishing environment that encourages more conceptual replication studies.
4. We believe adopting these practices will increase the confidence in results for field ecology. There are critical obstacles that could prevent some scientists from increasing within-study or between-study replication, including short-term funding mechanisms and the prospect of fewer publications. We suggest strategies to mitigate negative impacts to researchers, such as leading journals piloting new

article categories and explicit mention of experimentally linked studies. We acknowledge that adopting greater replication in field ecology will require significant changes to cultural practices, but there are clear benefits for improving our confidence in science.

KEYWORDS

confidence in science, experimental design, field ecology, open science, repeatability, replication, reproducibility crisis

1 | REPLICATION AND CONFIDENCE IN SCIENCE

Using best practices that enhance the reliability of research in ecological and environmental sciences is critical to generating confidence in scientific outcomes. The capacity to repeat experiments has long been a central tenant to the scientific process and has been described as what separates science from non-science (Dunlap, 1926; Popper, 1959, 1963). Replication has long been the hallmark of reliable science, and scientists have accordingly enjoyed public trust in many countries. However, recently there have been challenges in the media and the scientific literature surrounding a 'replication crisis' in science (Baker, 2016; Kelly, 2019). Included in public discussions are instances of apparent fraud (e.g. Pennisi, 2020; Simonsohn, 2013) and a decay in confidence and trust afforded scientists by the general public (e.g. Black & Rappuoli, 2010; Gauchat, 2012). Furthermore, the underlying causes are not simply differences in political orientation (Nisbet et al., 2015; Washburn & Skitka, 2017) and there are broader issues at play. There is much that we as scientists are unable to do in response to large socio-political pressures, but we can critically assess our scientific cultural habits and identify any changes that may increase confidence in our results.

A central challenge to all scientific disciplines is when multiple well-designed and conducted studies focused on the same research question yielding different results. This challenge is common across biological disciplines and typically indicates the initial findings may not have been representative of the true phenomenon (e.g. spurious result). Such outcomes are most likely to occur for highly novel and first-of-their-kind results, as there is not yet an established track record of independent validation. However, the complexities of the natural environment, physical limits of space and time, and changes in the living components of the study system (e.g. evolution, body condition, demographics) preclude recreating the exact same experimental conditions. Thus, we suggest field ecologists may be prone to assuming that discrepancies among studies are due to undescribed contingency, rather than the more parsimonious explanation that the initial (or subsequent) description of the phenomena was not accurate. A tendency towards discounting contradictory results, rather than the initial assertion of findings, limits understanding and reduces the ability of researchers and the public to have confidence in study outcomes.

The inability to rule out 'contingent factors' as a cause of contradictory results is not as prevalent in all biological disciplines. Biomedical experiments are frequently conducted in highly controlled environments utilizing identical laboratory materials (e.g. cell lines), reducing many of the contingencies that can mediate study outcomes. However, in field ecology, the likelihood of repeating outcomes declines with every change in implementation, including the research team, application of protocols, location, species composition, year/season and methods of analyses (Powers & Hampton, 2019). We term this the 'One Earth Problem', where working in a linear timeline on a non-replicated planet should cause inherently low rates of replicability for many aspects of field ecology. This raises the question: *What does a replication crisis in field ecology look like and what can we do about it?* While it may be unclear whether science in general (Cassey & Blackburn, 2006; Fanelli, 2009, 2018) or the discipline of ecology specifically (Fraser et al., 2020; Kelly, 2019) is in the midst of an actual replication crisis, examining approaches to improving confidence in field ecology is critical to ensure observed outcomes remain meaningful in this era of global change.

Field ecologists have tools available to enhance confidence in a study's results, and thus enhancing the reliability of research findings is in part within the control of individual scientists. We refer to these tools as sample replication, within-study replication and between-study replication (Table 1). We define *sample replication* as the number of independent units of observation/manipulation within the same statistical population (e.g. transects, focal animals, plots), where more sample replicates can increase sample size, improve precision and lower experimental error (Table 1). We define *within-study replication* as when a research team exactly repeats their own methods using the same materials in multiple locations or initiates the study in multiple years, all as part of a single study (Table 1). *Between-study replication* can be split into two: direct replication and conceptual replication. Direct replication is when a research team repeats a previous study under the same experimental conditions and using the same subjects and materials in identical conditions. Due to the One Earth Problem, direct replication is essentially impossible in field ecology, and instead what is more common will be conceptual replication. Conceptual replication is when a research team tests the same research question from a prior study, of their own or a different research group, using similar methods and materials, and under similar conditions. By relaxing some study parameters,

TABLE 1 A list of terms associated with practices and decisions that influence the resulting confidence of studies in field ecology. Here we use the term *study* to indicate a discrete scholarly unit (e.g. publication) that consists of a single research team (e.g. co-authors on a publication) answering a research question using a set of observational or manipulative experiments. We provide the definitions in this table to align with other research on replicability in science and provide consistency in discussions going forward within field ecology. There are unfortunately many overlapping terms that all begin with the letter 'r' and rather than invent new jargon, we use modifiers to clarify differing concepts

Term	Definition	Example
Within studies		
Precision	A measure of reliability among measurements taken within the same experiment and study	The standard error calculated from the biomass of multiple plants measured in a fertilization treatment
Repeatability	Similarity in measurements taken on the same individual or plot, under the same conditions, and within a short time period	Measuring the feeding rate of an individual bird by recording multiple feeding events within a feeding bout
Pseudo-replication	Treating experimental units as independent when they are not (<i>sensu</i> Hurlbert, 1984)	Applying fertilizer to a large plot and then treating subplots as independent replicates
Sample replication	Conducting the same observation on multiple independent units within the same statistical population (e.g. transects, target animals, plots)	The number of independently fertilized plots in a study of fertilization effects on plant diversity
Within-study replication	The same research team repeatedly initiating an experiment in multiple years and/or locations as part of a single study	Increasing the number of years where a fertilization addition experiment is tested on a plant community
Coordinated distributed networks	Experiments with standardized protocols run concurrently by multiple research groups in multiple locations, with results typically presented in a single study (see Fraser et al., 2013)	The same methodology of nutrient addition and plant measurement is applied at multiple sites globally, with each site operated by local researchers (e.g. NutNet, DroughtNet)
Among studies		
Direct replication	Repetition of a previous study using the same subjects and materials conducted under the exact same conditions. This form of replication is unlikely to occur in field ecology, but is more likely in other disciplines of biology	Using a controlled environment chamber set at a constant temperature, an experiment is run testing how a certain genotype of a species responds to fertilization. The experiment is later repeated using the same plant material and environmental conditions
Conceptual replication	Repetition of a previous study with the intention of testing the same underlying research question using similar methods within similar study systems. This is the most common type of replication in field ecology	After finding that another research team observed a decrease in grassland plant diversity with fertilization, a separate research team performs a similar test in their local grassland
Computational reproducibility	Identical analytical outcomes are obtained when an independent repetition of the analyses are performed using the same data, code and software as the original study	Wanting to better understand the results of a study which found that fertilization decreased plant diversity, a new research team downloaded the available data and code provided. In their own running of the analysis, they were able to repeat the original results
Meta-analysis	The quantitative synthesis of research results from multiple conceptual replication studies	While each independent study used in a meta-analysis may use different test subjects or ecosystems, these studies may be synthesized into a quantitative analysis because each shared the same underlying research question
Decision outcomes		
Sample size	The total number of independent replicates or observations within a study	
Generalization	The applicability of one's results beyond the individual study. The scope of generalization can include analogous time periods and sites, as well as changes to other conditions of the study design (e.g. species sampled, levels of treatments applied, frequency of sampling). This is largely a product of the inclusion of random effects, rather than fixed effects, for structural factors such as site (within-study replication) or variation in study outcomes (meta-analyses)	
Scientific reliability	The frequency that a study's results are held to be true when challenged with direct and conceptual replication efforts	
Scientific confidence	The belief in the results of a study that they are correct. Belief could come from satisfaction with the methods, trust in the researcher, demonstrated scientific reliability or anything that may or may not be scientific	

researchers use conceptual replication to repeat an experiment using the same hypothesis testing framework but using different methods or subjects (Fidler et al., 2017; Zwaan et al., 2018). We note that the difference between direct and conceptual replication is not binary but rather a continuum (for a discussion on categorizing replication studies see, Fraser et al., 2020; Nosek & Errington, 2017; Zwaan et al., 2018), though replication studies in field ecology will typically be closer to the conceptual end of that spectrum.

Direct replication is the clearest way to demonstrate a study's findings are robust, though this tool is largely absent from the field ecologist's toolkits. It is the inclusion of modest changes to conditions, methods and samples that largely limit our ability to know whether contradictory results indicate a spurious result or suggest interesting biology is at hand. Although this is unfortunate in the context of enhancing understanding of specific phenomena, there is a benefit. When studies with slightly different underlying conditions reach the same conclusion, we gain increased generalization; something which cannot come from direct replication studies.

Here, we discuss the issues of replication most directly relevant to field ecology, including the problems with a scientific culture focused on novelty over quality, tools to improve confidence in the scientific outcomes and ending with several suggestions for improving replicability in our discipline. We acknowledge that the ideas we are presenting are themselves not novel, but rather concerns and

sentiments raised frequently across disciplines in the scientific community in numerous forums and publications. While the focus of our review is on field ecology, there are other excellent examples present in psychology, behavioural science and medical sciences that we have done our best to include. However, this paper is not an exhaustive list of articles on the topic of replication in science, as that is better suited to fill several Ph.D. theses rather than a single journal article.

2 | PREFERENCE FOR NOVELTY OVER REPLICATION IN LEADING ECOLOGICAL JOURNALS

A brief survey of the Author Guidelines sections of high-impact ecology journals highlights that novelty is generally prioritized over reproducibility or replication (Table 2). Our journal sample represents the 12 of the top 20 journals (by impact factor) in ecology that primarily publish empirical field ecology studies (the remaining eight focus on reviews and modelling). On average, for every reference to replication, replicability or reproducibility, the Author Guidelines of these journals refer to aspects of novelty 6.7 times. Editorial decisions are not made exclusively based upon novelty, and all leading journals make efforts to ensure scientific rigour through the

TABLE 2 Frequency of synonymous terms of novelty and replicability in the author's guidelines section from 12 out of the top 20 ecology journals (by impact factor) that publish empirical field ecology studies. Synonyms for novelty included new, advance and advancing. Synonyms for replicability included replicate, replication, reproduce, repeat, reproducible and repeatable. To be counted, the term must refer to the content of the submitted manuscript. Terms used as adjectives of another were counted as one (e.g. new advances) and terms referencing the writing were excluded (e.g. new submissions, repeat figures)

Society or organization	Journal	Novelty	Replicability	Notes
British Ecological Society	<i>Journal of Applied Ecology</i>	5	0	
British Ecological Society	<i>Journal of Ecology</i>	7	0	
British Ecological Society	<i>Methods in Ecology and Evolution</i>	5	1	'Repeat' in the context of another researcher being able to repeat the methods
Ecological Society of America	<i>Ecological Monographs</i>	6	1	'Repeat' in the context of another researcher being able to repeat the methods
Ecological Society of America	<i>Frontiers in Ecology and the Environment</i>	7	0	
Independent	<i>Ecology Letters</i>	19	4	A Novelty Statement is explicitly required
Independent	<i>Global Change Biology</i>	6	0	
Independent	<i>Global Ecology and Biogeography</i>	4	0	
Independent	<i>Landscape and Urban Planning</i>	6	2	'Repeat' in the context of another researcher being able to repeat the methods
Nature	<i>Ecology and Evolution</i>	3	0	
Nordic Society Oikos	<i>Ecography</i>	6	0	
Society for Conservation Biology	<i>Conservation Biology</i>	6	4	Reproducibility in the context of another researcher being able to repeat the methods

peer-review process. However, this rigour does not translate into the same degree of explicit and repeated communication with prospective authors that are found for issues related to novelty. The expectation of novelty has been previously identified as a challenge for improving replicability in science (Smaldino & McElreath, 2016) as well as ecology and evolution (Parker, Forstmeier, et al., 2016). There is some evidence supporting a novelty over quality focus, with high-impact journals often publishing strong effects without better data quality relative to lower impact journals (Barto & Rillig, 2012).

An initial challenge in understanding replicability trends and practices is the rarity of between-study replication in the field ecology literature. There are many causes, though we note the evaluation structure of academia has turned towards frequently rewarding impact factors and related metrics (Archambault & Larivière, 2009). This metric-focused structure persists throughout much of the world despite its identified limitations and controversies (Alberts, 2013; Larivière et al., 2016; Vancley, 2012). One obvious constraint is that if the leading journals in field ecology do not require nor reward explicit replication efforts, there will be a disincentive within the community of scientists concerned with metrics of their resulting papers. Arguments in favour of greater between-study replication (Ihle et al., 2017) are often met with concern that it comes at the cost of personal success or the advancement of science (Hatchwell et al., 2017), highlighting the entanglement of these issues. The belief is that resources are finite and investing more into a single, extensively replicated study could come at a cost to many smaller studies potentially testing a wider range of hypotheses. There certainly are legitimate trade-offs in changing the status quo, though there is value in considering that as a discipline, field ecology may not be at the right balance.

Although novelty is highlighted in leading ecological journals and replication of experiments before publication is rarely required, this is not the case in leading journals of all biological sciences. In disciplines such as plant physiology and evolutionary genetics, direct replication is frequently performed as experiments are often conducted under highly controlled conditions (e.g. laboratory, greenhouse, growth chamber). The leading journals *Plant Physiology* (<http://www.plantphysiol.org/>) and *Cell Biology* (Cumming et al., 2007) require error bars to be calculated from repeated experiments rather than replicates within a single experimental trial. In the field-based discipline of agronomy, repetition of the experiment across multiple locations and years is not only recommended but it also should be maximized relative to the number of experimental replicates (Casler, 2015; McCann et al., 2012). These examples provide evidence that different cultural approaches are possible, even in field-based research.

Ironically, within-study replication is likely most critical for instilling scientific confidence for first-of-a-kind findings, as these studies are missing parallels in the literature and thus lack support from prior findings, increasing susceptibility to outlier results. To ensure confidence in research outcomes, cutting-edge research should bear the burdens of both being novel and demonstrating robust confidence in the results, which requires much more than sample size. One might

imagine that having a reputation for publishing non-replicable studies would harm a journal's impact factor. However, there is no clear self-correcting mechanism in a novelty-first publishing approach, particularly in a discipline that asks first 'I wonder what factor can connect the non-congruous results', rather than 'what evidence did the first study present that it was replicable?' Furthermore, even if many non-replicable studies do get dismissed by the scientific community, impact factors are generally driven by outlier papers of high citation rather than the long tails of infrequently cited work (Larivière et al., 2016; Seglen, 1997), potentially encouraging the pursuit of the occasional ground-breaking research from a publishing standpoint. A cultural and publication expectation that research teams demonstrate the replicability of their studies would certainly reduce the number of one-of-a-kind results, as some fraction of the results are necessarily spurious (i.e. we do not set significance equal to zero). This approach moves the burden of testing the validity of initial findings away from the larger scientific community (i.e. conceptual replication) and onto the original research team itself (i.e. within-study replication).

3 | REPETITION LEGITIMIZES

Independent of what is required by journal editorial boards, sample replication, within-study replication and between-study replication in field ecology can improve confidence in research outcomes. Here we describe the role of each in the context of field ecology.

3.1 | Sample replication

The goals of increasing sample replication are to improve the precision of the measured effect size, to provide adequate statistical power and to decrease the likelihood of a false estimate of effect size due to drawing numbers from a small pool of replicates. There have been multiple syntheses that have examined the role of replication in field ecology and addressed issues associated with low sample size, such as Type II errors—failure to reject the null hypothesis when it is false (Jennions & Møller, 2003), Type S errors—obtaining the wrong sign of the effect estimate (Parker, Forstmeier, et al., 2016) and Type M errors—observing an overestimated effect size (Lemoine et al., 2016). There has also been an extensive discussion in ecology concerning replication versus pseudo-replication (Colegrave & Ruxton, 2018; Hurlbert, 1984; Oksanen, 2001). Rather than repeat these important considerations of experimental design, we discuss a related issue, the relationship between effect size, error and replication.

We find preliminary evidence that greater within-study replication lowers the probability of observing a large effect size (Figure 1). Using two previous meta-analyses from the authors (Filazzola et al., 2019, 2020), we tested the reported sample size from each study against the calculated effect size (271 unique comparisons). Studies with relatively high sample replication ($n > 60$; Figure 1) rarely

report large effect sizes, whereas studies with few replicates ($n < 5$; Figure 1) have a wide distribution of effect sizes (see full workflow in Filazzola, 2021). There is no biological reason for this, as a researcher's degree of replication is not causally linked to the measured phenomena. Instead, the effect is likely statistical, where smaller sample sizes may include a greater proportion of outlier values relative to larger samples. Although this phenomenon is well known, its impacts on scientific confidence appear underappreciated. A combination of

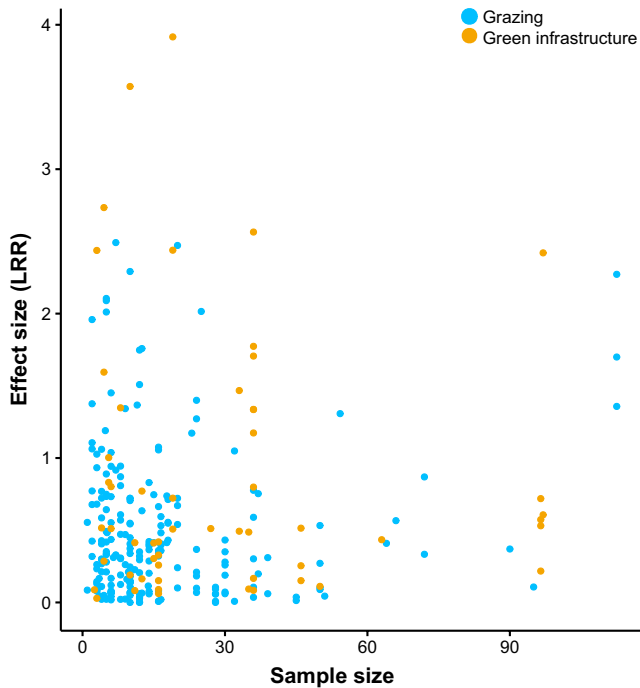


FIGURE 1 The relationship between sample size and the absolute effect size (log response ratio) estimated from studies within two meta-analyses (Filazzola et al., 2019, 2020)

large effect sizes and low sample size can be problematic, as it contributes to the perception of novelty (what a big effect!) but may simply be an outlier result, with the actual process much more modest (Forstmeier et al., 2017). Ironically, this statistical inevitability may cause a perverse incentive against high levels of replication in the pursuit of novelty.

Fortunately, there are simple actions that can be taken to reduce the influence of spurious outcomes in relatively poorly replicated studies. If the large effect size was due to stochasticity associated with random pulls from a small sample, then that would most likely be detected with even a minimum amount of within-study replication. Novel findings should thus be interpreted with caution when sample replication is low, and unfortunately there is no feasible a priori determination of what represents a low or high level of replication for a given phenomenon. In cases where study replication is not feasible, sensitivity and uncertainty analyses are both effective statistical tools for assessing the effects of sample replication and randomness of model output (Saltelli et al., 2004). We propose researchers explore the effect of systematically reducing sample size on the effect size of their outcome to determine whether their experimental design was under, over or adequately replicated (Figure 2). If removal of a few observations radically changes the effect size, there should be concerns raised about the confidence of the study's outcomes (Figure 2) and more sample replication should have been included in the design.

Though sample replication is critical for confidence in reported effect sizes, it does nothing to improve the generalization of findings beyond the study of interest nor does it provide any confidence that the entire study design was not confounded with unknown influences, that is, demonic intrusion (*sensu* Hurlbert, 1984). For these critical aspects of increasing confidence, the research team must turn to within-study replication.

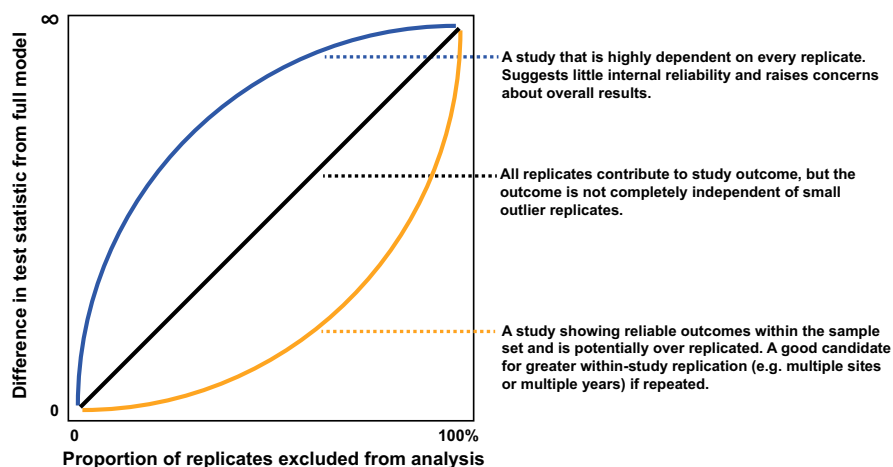


FIGURE 2 Removing a subset of the total replicates used in a study is expected to change the resulting test statistics (e.g. effect sizes, coefficients, R^2 , p values) relative to a statistical model where all replicates were included. Excluding no sample replicates would have no effect on the test statistics while excluding almost all the replicates is expected to have the maximum difference from the full model. Studies are likely to follow one of three general patterns with greater exclusion of replicates that determine where the experiment had high sample replication, whether there was an opportunity for greater within-study replication, and the overall confidence in the study design

3.2 | Within-study replication

Repeating experiments within a study can increase confidence that the observed pattern is biologically meaningful in broader environmental contexts, rather than one-off results. In field ecology, within-study replication can typically take two forms: (a) within a site, initiating the experiment in multiple time periods (e.g. years) and/or (b) within a single time period, repeating the experiment at multiple sites (Figure 3).

Despite the importance of replicating experiments to generate confidence in study outcomes, temporal replication is infrequently used in field ecology, with one review finding that it occurs in only 5% of multi-year studies (Vaughn & Young, 2010). However, inter-annual effects can have a pronounced effect on field ecology studies and the initiation year can alter the findings of a study (Vaughn & Young, 2010). Furthermore, this issue is not simply resolved by measuring the system for longer, as bias in initial conditions could persist indefinitely. For instance, initiating two different experiments before and after a volcanic eruption would never result in the same outcomes of a treatment on an ecological community regardless of how much time has elapsed.

Spatial replication may be possible when temporal replication is not, and it too enhances study reliability (when sites are nearly identical) and generalizability (when sites are similar but not nearly identical). We note that repeating experiments across multiple sites is distinct from running an experiment where multiple sites occur in different environmental conditions as part of the experimental design, such as studies testing the effects of environmental gradients. In this latter case, the number of sample replicates is the number of

sites used of each type, not the total number of sites. However, we are advocating for the repetition of entire experiments at similar sites without testing for additional variables as one means of enhancing scientific confidence. Repeating experiments in different sites can improve confidence that the study outcomes observed are accurate before research teams invest substantial time or resources, such as with a long-term study. For instance, site-selection bias in conservation biology has been attributed to identifying potentially artificial population declines in long-term studies (Fournier et al., 2019).

Addressing within-study replication is not a trivial undertaking, nor has it been completely ignored within field ecology. Some research groups address this challenge using coordinated distributed networks (e.g. NutNet, DroughtNet) where standardized protocols are applied across a wide range of locations (Fraser et al., 2013). Coordinated distributed networks are a form of within-study replication rather than between-study replication because individual sites are typically treated as replicates, rather than positions along a gradient. A strength of these approaches is the use of diverse field sites and initiation years that extends the statistical population of inference, and thus the generalizability of the results. Coordinated distributed networks have also received praise for being globally replicated science in a manner that is often not locally resource-intensive or demographically exclusionary (Borer et al., 2014; Stokstad, 2011). However, studies using coordinated distributed networks are not perfect. These experiments often have a lower sample size within site and year, lowering precision and accuracy at the site level. Despite standardized protocols, there can be variability among implementation, since these are different research groups conducting the same method. Additionally, the selection

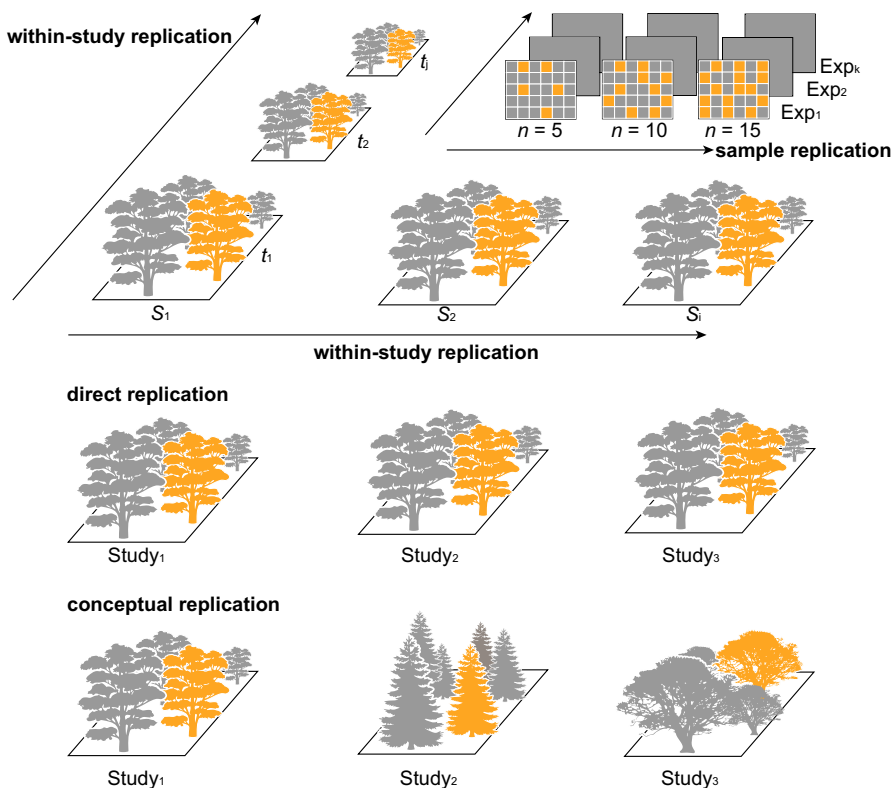


FIGURE 3 A framework for the different forms of replication in field ecology studies. In an experiment testing the response of trees to a treatment, researchers may increase sample replication to increase precision or accuracy of a single experiment. To help provide estimates of reliability of study outcomes, the research may also (1) repeat the experiment at different sites with nearly identical conditions (S_i) and/or (2) initiate the experiment over multiple time periods (t_j). Any differences among replicate sites and years of initiation will enhance generalization of the results by expanding the population of inference. Once completed, the study can be repeated either directly using the same materials and methods (i.e. direct replication) or conceptually using similar materials and methods (i.e. conceptual replication)

of standardized protocols requires careful consideration as small differences can be amplified across many sites and years. For example, non-congruent results describing the relationship between productivity and vascular plant diversity were obtained from two large coordinated networks (Adler et al., 2011; Fraser et al., 2015). Underlying these differences appears to be critical differences in sampling methodology due to differential interpretation of original theory (Fraser et al., 2015). While coordinated networks have great within-study replication, they too benefit from between-study replication efforts.

3.3 | Between-study replication

The vast majority of ecologists who responded to a survey have a positive view of direct replication studies concerning their importance, publication and funding (Fraser et al., 2020). Yet in a recent review, <0.1% of ecology studies are explicitly stated as replication studies of other research teams (Kelly, 2019). This failing of our discipline has been recognized, and there is broadly a call for increasing direct or conceptual replication in ecology (Fidler et al., 2017; Fraser et al., 2020; Kelly, 2019). Recognizing that one limitation to direct or conceptual replication has been that many journals explicitly preclude replication studies for publication requiring only *original* research articles (e.g. *Journal of Ecology*, *Ecology Letters*), several publishers recently launched journals designed to support replication experiments (*Experimental Results*, *ReScienceX*). It is unclear how contributions to these efforts will be weighted by both personal and institutional evaluative structures (e.g. hiring committees, tenure and promotion reviews, institutional rankings) in contrast to a novel-only approach.

Studies that are conceptually replicated form the core of the data used in ecological meta-analyses (Gurevitch et al., 2018; Stewart, 2010). In theory, meta-analyses should provide a robust estimate of the overall effect size of an ecological process/pattern, and through increasing the diversity of sites used they should increase generalization while also identifying clear contingencies. Advances in meta-analytic techniques, such as using random effects to estimate multiple populations within the dataset, have improved confidence in estimating a generalizable pattern. However, both systematic reviews and meta-analysis are not without problems in their approach including publication bias, incorrect analyses and statistical violations (Haddaway et al., 2020; Koricheva & Gurevitch, 2014; Spake & Doncaster, 2017). Most critical to this discussion are concerns that these syntheses are drawing together studies that are not from equivalent populations due to meaningful differences in protocols and implementation, or pseudo-replicated using non-independent data such as those that are nested within the same study, research group or location (Ioannidis, 2016; Koricheva & Gurevitch, 2014; Whittaker, 2010). Greater clarity around the core hypothesis being tested within each study could assist in identifying groupings of parallel studies and help untangle contingencies driving heterogeneity in the observed pattern.

Problems begin to emerge with meta-analyses when multiple studies differ widely in outcomes, which is expected if the pattern has a small effect size with high variability, or some contingent factor is missing from the analysis. Just as few investors investigate every stock held in a mutual fund, reviewers and readers of meta-analyses do not typically assess the suitability of every study used within an analysis. Instead, we assume that authors of the meta-analysis did not lump together distinct studies nor embed pseudo-replication by including studies by the same research team or from the same study location. In some cases, reproducing a meta-analysis using the same target studies has produced different results than the original [(Maestre et al., 2005) versus (Lortie & Callaway, 2006), or (Serrouya et al., 2019) versus (Harding et al., 2020)]. While meta-analyses remain a powerful synthetic tool, they should not be relied on as a solution to the lack of within-study replication or between-study replication.

4 | SUGGESTIONS FOR IMPROVING CONFIDENCE IN FIELD ECOLOGY

4.1 | Increase within-study replication

We suggest the first step for improving confidence in study outcomes in field ecology is to put greater emphasis on within-study replication. We recognize project length and the number of project locations can be limited by hard restrictions (e.g. granting periods, graduate student programmes) and 'softer' restrictions (e.g. money, labour), but often the limitations may be out of habit or intentionally targeting the least publishable unit (Leimu & Koricheva, 2005; Rau et al., 2017). Within a research team, replication over time and space could be accomplished with similar resources by sacrificing some sample replication or using simpler designs that omit a treatment. Among research teams, coordinated distributed networks offer an accessible format for participation in widely repeated experiments. To date, such networks tend to be global, which is critical for assessing some research questions. However, the use of smaller localized networks would also go a long way towards increasing confidence in study outcomes. Increasing participation among institutions and laboratories provides an opportunity for shared resources to repeat experiments.

Redesigning an experiment could generate instances of within-study repetition while having a small impact on sample replication. The asymptotic nature of test-statistic distributions identifies diminishing returns for increasing precision with increasing sample size. Thus, sacrificing some sample replication in favour of within-study replication could benefit the generalizability of the findings with minimal impact on statistical confidence. Researchers could explore the sensitivity of their study outcomes to determine whether removing a fraction of observations would have little effect on the findings, thus potentially well suited for revising the design to increase within-study replication (Figure 2). Many statistical methods also allow for the combination of small experiments into a more powerful

outcome than using the replication from each experiment, such as Fisher's combined probability test (Fisher, 1992) or z-transformation test (Stouffer et al., 1949). With equivalent resources, one can design an experiment that is not too onerous yet still accomplishes some degree of within-study replication.

4.2 | Increase reproducibility for aspects that we can control

Although many aspects of field studies cannot be reproduced by other groups, some aspects of these studies can be exactly replicated. Much of the ecology literature to date on reproducibility focuses on increasing access to data (Hampton et al., 2013; Reichman et al., 2011), code availability (Culina et al., 2020; Powers & Hampton, 2019; Ram, 2013), pre-registering experiments (Parker, Forstmeier, et al., 2016; Parker et al., 2016) and experimental design transparency (Parker, Nakagawa, et al., 2016). With the recent popularity of computational ecology methods (e.g. species distribution modelling, climate change scenarios), there have been calls to assist reproducibility efforts through greater code sharing (Culina et al., 2020), more reusable open data (Roche et al., 2015) and machine-readable meta-data (Poisot et al., 2019; Wickham, 2011). An important element of this 'computational reproducibility' includes the repetition of the analyses which requires access to the raw data acquired and robust meta-data. Before conducting fieldwork, pre-registration of experiments can be effective in defining clear questions and hypotheses to be tested free of bias, and provide greater credibility to the produced trends (Nosek et al., 2018). Many of these tools have the goal of minimizing questionable research practices and improving our confidence in science. Recently, the Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology—SORTEE (<https://www.sortee.org/>) was established to address these aspects of reproducibility. Discussing each of these aspects here would be beyond the scope of this review, but we encourage such efforts of reproducibility and support the goals of SORTEE.

4.3 | Divest from novelty and invest in design

Advancing ecological research is important, but we may be collectively too focused on novelty and originality at the expense of estimating reliability and generating confidence in scientific outcomes (Forstmeier et al., 2017; Nosek et al., 2012; Smaldino & McElreath, 2016). The pursuit of knowledge motivates most practicing scientists, and there are professional awards associated with publishing in certain venues (Parker, Forstmeier, et al., 2016). However, the current focus on novelty encourages a mindset to pursue 'new' rather than lend confidence to that which is already thought to be true. We do not suggest that individual scientists need to be martyrs to the cause and sacrifice professional development to increase replication in science. Instead, some practices can be adopted across

the community that can improve quality without being to the detriment of individuals' careers.

For publishers, explicit statements about study reliability are needed and should be considered during manuscript evaluation. This need not come at the cost of novelty, but instead the bar of demonstrating novel should also include demonstrating reliability. We encourage editorial boards and leadership groups to take the lead provided by other biological disciplines and more fully incorporate expectations of replication into the author's guidelines and instructions to authors. Journals could be more explicit about what they consider to be a well-conducted study, including the elements of sample replication, within-study replication, conceptual replication and computational reproducibility, and make this clear both to authors and reviewers (see the Transparency and Openness Promotion Guidelines; Nosek et al., 2015). We are confident that if the leading journals continue to publish important and novel work, while also increasing the expected standards for generating confidence, scientists will respond accordingly.

Another approach could include creating a *Replicability Section* either within the manuscript or published data that identifies similarities with other research to assist future meta-analyses and replication studies. If not in an independent section of the manuscript, embedded in the discussion ecologists can explicitly define the elements of their design that would be most useful to be replicated in future studies. This can help identify opportunities for replication from the authors most closely connected to the research area. Furthermore, if the authors conducted a pilot study, they could submit these results as direct or conceptual replication and be described in this replicability section. Finally, the section can function as an indicator to reviewers as to the potential replicability of the study. A cultural adoption (or through editorial policy) of studies explicitly stating previous work that is being leveraged would also do well to identify the instances of conceptual replication that already exist, but that may be intentionally obfuscated to emphasize novelty.

4.4 | Promote conceptual replication studies

A low frequency of explicitly indicated replication studies in field ecology is not consistent with the proliferation of meta-analyses in the discipline. It is possible that authors of original empirical studies are intentionally minimizing any perceived similarity to other published studies as a way of highlighting novelty. We suggest that if conceptual replication efforts were celebrated and the emphasis on novelty moderated, it would free authors to be more explicit about potential causes of similarity or differences with other studies. This would include a cultural adoption that replication studies have value in funding competitions, professional reviews and job applications.

Only through conceptual replication can we validate the reliability of results and identify potentially unreliable results in field ecology, which is an essential component of science. One strategy to improve this balance can include piloting a new category of articles at leading ecology journals dedicated to between-study replication where

the expectation is repeating another researcher team's experiment and no expectation of novelty (Parker & Nakagawa, 2017). A similar practice currently exists within the Association for Psychological Sciences where replication studies use a different peer-review process. Another strategy could be that a journal supports a no-page charge online repository of citable studies that have been specifically identified to be conceptual replications of published articles, assuming it meets some standards. In this approach, the journals would bear some responsibility in trying to coordinate replication studies of their own published catalogue. Finally, ecology societies or leading publishers could develop a completely independent and low-cost journal on replication studies to fit within their family of journals.

4.5 | Be sceptical of contradictory findings

In ecology, we have adopted a tradition of assuming differences in outcomes are due to the complexity of nature, and thus we accumulate what has been described as 'zombie ideas' (Fox, 2013; Moles & Ollerton, 2016). For example, the Intermediate Disturbance Hypothesis remains a textbook ecology concept despite independent studies and a meta-analysis demonstrating that a unimodal relationship between species diversity and the frequency of disturbance occurs infrequently (Fox, 2013; Mackey & Currie, 2001). Our discipline appears to focus on potential uncontrolled variation as causal to differences in outcome, which may be true, but it should be demonstrated not assumed. We suggest that we should adopt a different mindset: If conceptual reproducibility efforts do not support original works and no causal contingency has been clearly demonstrated (not just suggested), then the original findings likely are not representative of the natural process and should be treated cautiously (Nakagawa & Parker, 2015). That is not to say that previous concepts, such as IDH be omitted from ecology, but rather enough evidence has accumulated that a revision is needed based on advanced understanding.

5 | IMPACTS ON SCIENCE AND SCIENTISTS

The goal of our paper is not to generate a cadre of researchers dedicated to debunking or confirming previous studies, but rather to emphasize the significant imbalance in weighing novelty over scientific rigour. Creating a generation of scientists committed to only repeat studies can be equally as undesirable as a generation where repeat studies are non-existent, but small steps exist to achieve greater replication. Increasing within-study and between-study replication in field ecology will not be a simple change, but it could improve our confidence in scientific outcomes. While this may reduce the total number of papers produced (because resources, funding and labour are finite), that is not necessarily an unsolvable issue or even an actual problem.

There are several potential strategies to mitigate the potential for reduced publications, particularly among early-career scientists. For example, one could design more replication experiments that use either a two-step or hybrid design to reproduce a previous experiment while also expanding with new method refinement or additional contingencies identified. In this model, there would be intentionally more overlap in the intellectual space across studies, though not formally designating studies as repeat or novel. These types of studies may also be more eligible to receive funding under the current expectations of novelty because the experiment would be designed in a manner that could be compared to previous studies and include new considerations to identify additional contingencies. A second strategy could be an expanded use of coordinated distributed experiments, both locally and globally. Network approaches can be a cost-effective way of doing research, and can result in substantial publication rates for active team members.

Alternatively, we need not assume lower publications rates are an actual problem, if the scientific community agrees to value contributions to both novelty and repeatability. For example, other disciplines (e.g. physiology, cell biology, agronomy) have adopted requirements for greater within-study replication and are equally credible. Within agronomy, there is a focus on maximizing within-study replication at the cost of sample replication (Casler, 2015; McCann et al., 2012). In ecology, the key for this to be successful is for those in leadership and decision-making positions (editors, granting agencies, tenure and promotion evaluation) to emphasize the value of these efforts, and recognize maximization of publication numbers is not the goal of the scientific enterprise. Efforts should be led by those able to absorb change (i.e. more senior scientists). We will need to be cautious that within-study replication is not used as a form of gatekeeping to marginalize some research groups. However, it is important to note that the desire for 'novelty' is often used for gatekeeping (Siler et al., 2015), and perhaps we should adopt criteria surrounding replication that are more quantifiable than novelty. It will be difficult for a scientific culture to change and there will be resistance, but after a few generations of scientists a new culture may be adopted and confidence in findings improved.

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AUTHORS' CONTRIBUTIONS

A.F. and J.F.C. contributed equally to the manuscript including the conception of ideas, design, interpreting the results, drafting the

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PEER REVIEW

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DATA AVAILABILITY STATEMENT

Data used for comparison of effect size and replication can be found at Filazzola et al., 2020: <https://doi.org/10.6084/m9.figshare.12075009.v1> and Filazzola et al., 2019: <https://doi.org/10.5063/F1S180S8>. All code and analyses used are publicly available (Filazzola, 2021: <https://afilazzola.github.io/SpaceVTime/>).

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